Mechanistic Interpretability of Vision Transformers: Sparse Autoencoder Feature Decomposition for Visual Concept Discovery

Vito Ferri
Politecnico di Torino
Turin, Italy

Fabio Mastromauro
Politecnico di Torino
Turin, Italy

Marco Ravaioli
Politecnico di Torino
Turin, Italy

ABSTRACT

We train Sparse Autoencoders (SAEs) on the MLP output activations of a pre-trained ViT-B/16 at two network depths, and use CLIP as an external cross-modal evaluator to assign interpretable labels to discovered sparse directions—a strategy designed to address the labelling gap that arises in purely visual transformers trained without text supervision. Layer 6 features encode low-level texture and geometric patterns with negligible causal impact (mean relative logit drop: 0.03%); Layer 11 features encompass more semantic concepts and are substantially more causal (mean: 1.40%), concentrated on a single feature (*honeycomb pattern*, 13.49% logit drop under ablation). These results confirm that SAE-based decomposition transfers to purely visual transformers and that external VLM evaluation is a viable labelling strategy.

KEYWORDS

Mechanistic Interpretability, Vision Transformers, Sparse Autoencoders, Polysemanticity, Feature Decomposition, Explainable AI

1 INTRODUCTION

As neural networks are deployed in critical settings, mechanistic interpretability aims to reverse-engineer their computations into analyzable units [6]. Sparse Autoencoders (SAEs) have proven effective at decomposing polysemantic MLP activations of language models into monosemantic features [1, 4]. Applying this to Vision Transformers (ViTs) [5] introduces two specific challenges absent in the language setting: the spatial structure of patch-based attention makes causal tracing harder to formalise, and—crucially—ViTs trained without language supervision provide no built-in mechanism for translating sparse visual directions into human-readable concept labels. This project addresses both challenges by training SAEs on a ViT-B/16 and using CLIP [9] as an external evaluator for cross-modal concept labelling, providing the first systematic evidence that mechanistic interpretability tools transfer to purely visual transformers.

2 RELATED WORK

2.1 Mechanistic Interpretability in Transformers

Elhage et al. [6] formalised the residual stream framework: the residual stream acts as a central communication channel, with attention heads routing information between tokens and MLP blocks processing within-token representations. Understanding this residual communication is crucial for identifying which layers and components drive specific model behaviours. Building on this, Conmy et al. [3] introduced ACDC (Automated Circuit Discovery), which identifies the minimal computational subgraph responsible for a target behaviour via iterative activation patching—establishing a rigorous methodology for causal analysis that is, in principle, architecture-agnostic.

2.2 Sparse Dictionary Learning and Monosemanticity

A significant obstacle in mechanistic interpretability is *polysemanticity*—individual neurons activating in response to multiple, unrelated features—attributed to *superposition*, where models encode more features than they have dimensions [6]. SAEs applied to MLP activations or residual stream vectors learn a larger dictionary of sparse, monosemantic features that influence model predictions in predictable ways [1, 4]. These contributions target language models exclusively: tokens are words, features correspond to linguistic concepts, and verifying interpretability amounts to reading the activating tokens.

2.3 Internal Representations of Vision Transformers

Raghu et al. [10] showed that ViTs aggregate global information early via self-attention and propagate features strongly through residual connections—properties with no direct analogue in language models that complicate circuit-level analysis. DINO [2] further demonstrates that training the same ViT architecture through self-distillation, without any label supervision, gives rise to qualitatively different emergent properties—most notably, attention maps that spontaneously segment salient objects—suggesting that the character of sparse features discovered by SAEs may depend substantially on the training regime.

2.4 Concept Discovery in Vision-Language Models

Gandelsman et al. [7] decompose CLIP image representations as sparse sums of text-direction embeddings, yielding human-interpretable visual decompositions. Haque et al. [8] apply SAEs to medical VLMs, anchoring discovered concepts to clinical ontologies and evaluating them with an LLM as external judge. Both approaches rest on pre-existing text-image alignment: the visual concepts are interpretable precisely because they live in a space where text directions are meaningful. In a ViT trained without language supervision, no such space exists—and it is exactly this challenge that this project addresses.

3 RESEARCH GAPS

Three gaps motivate this work. **Systematic SAE analysis for pure vision models**: SAEs have not been applied to ViTs trained without language supervision; visual tokens encode image patches rather than words, and there is no direct way to extract a human-understandable label from a learned sparse direction—making it an open empirical question whether SAEs recover meaningful structure from visual activations. **Spatial structure of ViT attention**: unlike language transformers, ViTs operate over all patch pairs simultaneously with no preferred direction of information flow [10]; tracing how evidence from spatial regions is compressed into the [CLS] token requires causal tools not yet validated in the visual setting. **Cross-modal labelling**: when no internal text-image alignment exists, assigning interpretable labels to SAE features requires an external grounding mechanism—a bottleneck unacknowledged in most vision interpretability work. These gaps converge on one open question: can SAEs recover interpretable, monosemantic features from a purely visual ViT, and how can those features be evaluated without built-in language supervision? This project directly targets that question by using CLIP [9] as a detached external evaluator, decoupling the interpretability analysis from the backbone under study and avoiding any need for a vision-language architecture.

4 METHODOLOGY

Our pipeline has four stages.

Backbone and data. We analyse ViT-B/16 [5] (12 transformer blocks, d=768, trained on ImageNet-1k) using 1,000 validation images paired with an 18-concept CLIP vocabulary spanning object categories (*animal*, *bird*, *plant*, ...), textures (*fur texture*, *feather texture*, ...), and geometric patterns (*spotted*, *honeycomb*, *scale*, ...).

Activation extraction. Forward hooks collect MLP output activations at Layer 6 (mid-network) and Layer 11 (late-network). Each image yields a 196×768 patch-token matrix per layer ([CLS] excluded), centred by the dataset-level mean before SAE training.

Sparse Autoencoder. Given activation $\mathbf{x} \in \mathbb{R}^d$, the SAE produces a sparse code

$$\mathbf{z} = \mathrm{ReLU}(\mathbf{W}_{\text{enc}}\mathbf{x} + \mathbf{b}_{\text{enc}}), \quad \mathbf{z} \in \mathbb{R}^m, \tag{1}$$

and reconstructs $\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}}\mathbf{z}$ (unit-norm columns, m=8d=6144). Training minimises

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda\|\mathbf{z}\|_1 \tag{2}$$

(Adam, lr$=10^{-3}$, $\lambda = 10^{-3}$, batch 256; 5 epochs for Layer 6, 15 for Layer 11).

CLIP labelling. The K=5 highest-activating crops per feature are matched to the vocabulary via mean cosine similarity to *"a photo of {concept}"* prompts [9]. A feature is characterised as monosemantic when the CLIP-assigned label is confirmed by visual inspection of the top-activating crops.

Causal verification. Feature k is ablated by subtracting its contribution from patch-token activations: $\mathbf{x}' = \mathbf{x} - f_k(\mathbf{x})\,\mathbf{W}_{\text{dec}}[:, k]$.

Table 1: SAE metrics and mean causal impact. $\bar{L}_0$: avg. active features per patch token.

Layer	R^2	$\bar{L}_0$	Mean RLD
6 (mid)	0.9957	201.72	0.03%
11 (late)	0.9477	107.90	1.40%

Causal impact is quantified via the Relative Logit Drop (RLD):

$$\mathrm{RLD} = \frac{L_{\text{baseline}} - L_{\text{ablated}}}{|L_{\text{baseline}}|}. \tag{3}$$

A dose-response sweep over ablation strength $s \in [0, 1]$ and a steering experiment (s=5, amplifying the feature's contribution) together rule out artefactual interpretations of the causal signal.

5 EXPERIMENTS AND ANALYSIS

Table 1 reports SAE quality and mean causal impact per layer.

Layer 6 achieves near-perfect reconstruction (3.3% activation density); its mean logit drop of 0.03% confirms that mid-network features are individually causally negligible. Layer 11 has sparser activations (1.8% density) and a mean logit drop 44× higher, concentrated on a single feature. Table 2 lists all ten features per layer.

5.1 Feature Variation Across Depths

Layer 6 features are dominated by low-level texture and geometric patterns (*spotted pattern*: 4 features; *scale pattern*: 2; plus *honeycomb pattern*, *fur texture*, *furniture*, *pineapple*), consistent with the role of mid-network MLP layers as local structure detectors [10]. At Layer 11, the vocabulary shifts towards semantically abstract concepts (*animal*: 2 features; *plant*: 3; *bird*), alongside texture descriptors. Visual exemplars for the top-5 Layer 11 features are shown in Figure 1.

5.2 Causal Analysis

Feature 5065 at Layer 11 (*honeycomb pattern*) accounts for 13.49% relative logit drop under ablation and 5.68% logit increase under amplification; all remaining nine Layer 11 features show drops below 0.4%. The spatial heatmap and dose-response curve for Feature 5065 are shown in Figure 2: the heatmap confirms selective activation on coherent image regions, and the monotonically increasing dose-response curve rules out artefactual interpretation.

5.3 Discussion

The results address both research gaps. SAEs recover structured, labellable features from purely visual MLP activations: all features at both depths receive distinct CLIP labels, confirmed by visual inspection of top-activating crops. Human assessment independently validated CLIP labels for texture-dominant features; convergence on Feature 5065—from both CLIP scoring and human review, combined with the 13.49% logit drop—constitutes the strongest evidence of monosemantic, causally relevant behaviour. The depth-wise shift from texture detectors (Layer 6) to semantic features (Layer 11) reproduces the representational hierarchy expected from prior ViT analyses [10]. The concentration of causal impact on a single late-network feature may reflect specialisation in the [CLS]

Table 2: Ten most active features per layer. CLIP score: mean cosine similarity to top-K crops. RLD: relative logit drop under full ablation (s=0). Steer: logit change under 5× amplification.

Layer	Feature	CLIP concept	CLIP score	RLD (%)	Steer (%)
6	4770	spotted pattern	0.2669	+0.04	+0.16
6	4271	honeycomb pattern	0.2651	−0.04	−0.03
6	4506	scale pattern	0.2536	+0.69	+1.78
6	2719	spotted pattern	0.2511	+0.56	+0.80
6	2045	pineapple	0.2487	+0.03	+0.11
6	4608	fur texture	0.2472	+0.29	−0.01
6	5536	spotted pattern	0.2481	−0.10	−1.20
6	2041	furniture	0.2476	+0.06	+0.17
6	1632	spotted pattern	0.2454	−1.21	−4.64
6	4343	scale pattern	0.2344	−0.00	−0.00
11	**5065**	**honeycomb pattern**	**0.2970**	**+13.49**	**+5.68**
11	3932	fur texture	0.2794	+0.00	+0.00
11	965	plant	0.2683	+0.00	+0.00
11	540	striped pattern	0.2578	+0.00	+0.02
11	1767	animal	0.2571	+0.02	+0.03
11	5517	pineapple	0.2477	+0.36	+0.50
11	4488	plant	0.2505	+0.00	+0.00
11	2838	animal	0.2339	+0.05	+0.15
11	3591	plant	0.2395	+0.11	+0.10
11	5771	bird	0.2508	−0.00	−0.00

accumulation mechanism, though a larger image sample is needed to confirm whether this pattern is input-specific or systematic.

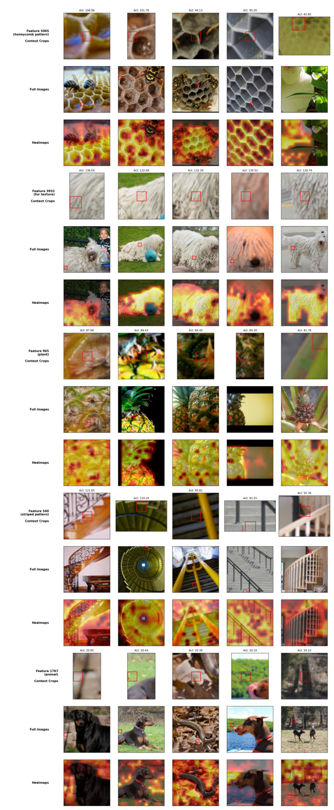

Figure 1: Top-5 Layer 11 features (by CLIP confidence). Each row: contextual crops, full images with highlighted patch, activation heatmaps. Feature 5065 (*honeycomb pattern*) exhibits the highest causal impact (13.49% RLD).

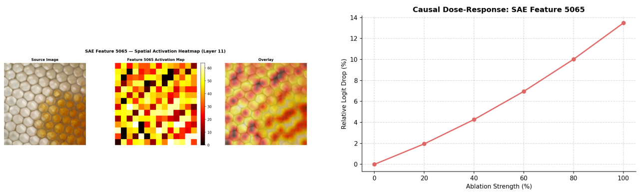

Figure 2: Feature 5065 (Layer 11). Left: spatial activation heatmap showing selective activation on coherent image regions. Right: dose-response curve—monotonically increasing logit drop as ablation strength increases from 0 to 1.

6 CONCLUSIONS

SAE-based feature decomposition transfers to purely visual Vision Transformers: SAEs trained on ViT-B/16 MLP activations recover sparse, interpretable features at two depths, and CLIP-based external evaluation provides a viable labelling strategy without internal text supervision. Mid-network features (Layer 6) are dominated by texture detectors with negligible causal footprint; late-network features (Layer 11) encompass more semantic concepts, with causal impact concentrated on a single highly specialised feature (Feature 5065, 13.49% logit drop). Limitations include the finite and ImageNet-biased CLIP vocabulary, the restriction to MLP outputs (attention heads and residual stream remain unexplored), and the qualitative nature of the monosemanticity assessment. Natural extensions include circuit-level activation patching to trace classification-specific computation, and a comparative analysis of SAE dictionaries from supervised versus DINO-pretrained ViTs [2]—two regimes known to produce qualitatively different internal representations [10].

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